

From Documentation to Infrastructure: The Development of 3D Cultural Heritage Digitization in Ukraine

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Abstract

The analytical article highlights the current state and stages of 3D digitization of cultural heritage in Ukraine, examining in depth technological practices such as lidar and photogrammetric scanning, 3D manufacturing, and their application in digital products, including interactive projects, games, museum exhibitions, and AR/VR applications. The research addresses the question: What technical, organizational, and methodological factors currently limit the development of FAIR 3D cultural heritage infrastructures in Ukraine, and what approaches can improve their sustainability?

The aim of the study was to survey the transformation of the 3D digital content production industry in Ukraine, to analyze the current state of the long-term preservation infrastructure and its online publication. The study investigates how the Ukrainian 3D cultural heritage community is evolving from isolated documentation initiatives towards a sustainable digital preservation infrastructure under wartime conditions.

The object of the study was the Ukrainian professional digital community, which includes specialists in 3D scanning and documentation of cultural heritage, their implemented projects, and published data.

The survey methodology was developed by the authors based on the educational framework of the 3D-4CH consortium, which categorizes the 3D cultural heritage sector into distinct stakeholder groups, including policymakers, managers, and practitioners. This categorization enabled the questionnaire to address role-specific competencies, responsibilities, and challenges, allowing the identification of the specific needs, knowledge gaps, and practical experience of each respondent group. The survey results therefore provide a differentiated understanding of the current state of the Ukrainian 3D

cultural heritage community and inform the proposed infrastructure and capacity-building recommendations.

Data were collected through semi-structured interviews with Ukrainian cultural heritage professionals and an online questionnaire completed by 60 respondents representing policymakers, managers, practitioners, and the wider cultural heritage community, analysed using thematic qualitative analysis.

The key strategic goal for industry experts is the integration of cultural heritage data into international catalogues such as Europeana, through interoperable metadata, standardized documentation workflows, and FAIR-oriented data management practices that support long-term preservation, and reuse, which should ensure the security and accessibility of digital copies of Ukrainian cultural heritage, especially those under threat of destruction, and facilitate research and interpretation on a worldwide scale. The study identified systemic challenges related to the limited technical infrastructure, lack of financial and human resources, standards for the preservation and publication of 3D data, and insufficient coordination between the public and private sectors, which significantly affect the results of the work.

The proposed infrastructure model connects 3D acquisition, metadata creation, repository-based preservation, online publication through interoperable web-based workflows compatible with European digital heritage ecosystems.

The speed of change in the industry significantly exceeds the pace of adaptation of the state's institutional policy and initiatives, which complicates the formation of a sustainable ecosystem for preservation. The authors emphasize the need to move from fragmented initiatives to a holistic infrastructure model that combines the processes of creation, preservation, and reuse of 3D cultural heritage data.

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